Zainab KHALLOUF

PhD in Computer Science
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SUMMARY

With over 20 years of experience in academic teaching, I have developed a deep belief in the transformative power of education. I am currently seeking a permanent academic teaching and research position that aligns with my skills, experience, and long-term aspirations. Since 15 Nov 2021, I'm a lecturer-researcher (CDD-LRU) at Université Bretagne Sud (UBS) and before that, I was a lecturer at the Faculty of Informatics Engineering-Homs-Syria from fall 2006 to oct 2021. My teaching portfolio includes a broad range of courses such as blockchain, component-based software engineering, computer networks security, E-commerce systems, distributed systems, and computer ethics. I hold a Ph.D. in Informatics from Grenoble Institute of Technology (INPG), France, with dissertation entitled: "Multicast Routing Infrastructure Security: The Network Operator's Point of View", supported by a CIFRE scholarship. My research expertise bridges academia and industry, having collaborated with France Telecom RD and INRIA Grenoble. Prior to that, I received a Master's Degree: DEA ISC (Diplôme d'Etudes Approfondies en "Informatique : Systèmes et Communications") with dissertation entitled: "A Study of High Speed Forward Error Correction (FEC) Codes for the Reliable Multicast Delivery of Bulk Data" in 2002 and Postgraduate Diploma in Computer Science: DESS-CCI (Diplôme d'Etudes Supérieures Spécialisées Compétence Complémentaire en Informatique) in 2001 from Joseph Fourier University, Grenoble, France. My initial training was in Syria at Al-Baath university (Homs city) where I received my bachelor's degree in mathematics specializing in Computer Science in 1997, and Postgraduate Diploma in Computer Science in 1999. Because I graduated as the top of my class with an average grade of 80.2/100, I was appointed as a demonstrator in the university and I was later awarded a scholarship from the ministry of higher education-Syria, allowing me to continue my higher studies in France.

EDUCATION AND QUALIFICATIONS

Dec. 2002 – Mar. 2006 **PhD in Computer Science – Systems and Software**

Grenoble Institute of Technology (INPG), France

PhD funded by the French National Association for Research and Technology (ANRT) through a CIFRE industrial research agreement, in collaboration with France Télécom R&D (Lannion) and INRIA Rhône-Alpes (Grenoble).
Thesis: Multicast Routing Infrastructure Security: The Network Operator's Point of View.
Examination Committee: Prof. Guy Mazaré (ENSIMAG), Chair; Dr. Isabelle Chrisment (LORIA, Université Henri Poincaré – Nancy 1), External Examiner; Prof. Bernard Cousin (IRISA, University of Rennes), External Examiner; Dr. Olivier Paul (INT, Évry), Thesis Director; Prof. Andrzej Duda (LSR-IMAG, ENSIMAG), PhD Supervisor; Dr. Vincent Roca (INRIA Rhône-Alpes), Co-supervisor; Sébastien Loye (France Télécom R&D), Industrial Supervisor.

Sept. 2001 – July 2002 **DEA (Postgraduate Diploma) in Computer Science – Systems and Communication**

Joseph Fourier University, Grenoble, France

Dissertation: A Study of High-Speed Forward Error Correction (FEC) Codes for the Reliable Multicast Delivery of Bulk Data.

Sept. 2000 – July 2001 **DESS (Postgraduate Professional Diploma) in Computer Science**

Joseph Fourier University, Grenoble, France

Additional postgraduate qualification in computer science.

Sept. 1999 – June 2000 **Intensive French Language Course**

Damascus, Syria

Sept. 1997 – Aug. 1999 **Postgraduate Diploma in Computer Science**

Al-Baath University, Homs, Syria

Final-year team project: Using Unicode and Java to Solve the Problems of Arabic Language in Microsoft Internet Explorer.
Grade: Very Good – average: 76.67/100.

Sept. 1993 – June 1997 **Bachelor's Degree in Mathematics, Computer Science Specialization**

Al-Baath University, Homs, Syria

Three years of applied mathematics and one year of computer science.
Grade: Excellent – average: 80.2/100 (*Top Student Award*).

PROFESSIONAL EXPERIENCE

France – ENSIBS, Université Bretagne Sud

Since Sept. 2025 **Full-time Fixed-term Contract Lecturer (CDD-LRU)**

ENSIBS, Université Bretagne Sud

Nov. 2021 – Sept. 2025 **Full-time Fixed-term Contract Lecturer and Researcher (CDD-LRU)**

ENSIBS, Université Bretagne Sud

Syria – University of Homs, Faculty of Computer Engineering

Dec. 2015 – Oct. 2021 **Assistant Professor**

Vice-Dean for Scientific Affairs (August 2017 – August 2018)

Sept. 2006 – Dec. 2015 **Full-time Lecturer and Researcher**

Faculty of Computer Engineering

Sept. 2000 – May 2006 **Study Leave in France**

Franco-Syrian Scholarship

Oct. 1998 – Aug. 2000 **Full-time Teaching Assistant**

Faculty of Science (Department of Mathematics, specializing in Computer Networks), followed by the Faculty of Computer Engineering (from January 1999)

TEACHING EXPERIENCE (OVERVIEW)

ENSIBS – Université Bretagne Sud (since 2021). Approximately 1,824 teaching-equivalent hours (HETD) from 15 November 2021 to June 2026. Teaching at engineering degree level (Cybersecurity and Data Science, Software Cybersecurity) and in the preparatory programme. Subjects include web application security, blockchain and Web3, statistical and neuro-symbolic AI for cybersecurity, distributed and massively parallel computing, secure development of distributed systems, component-based software engineering, behavioural analysis, and programming. Supervision of internships, final-year projects, and work-study students.

University of Homs (Syria), 2006–2021. More than 3,000 teaching hours, mainly in network security, information security, distributed systems, and computer networks, at undergraduate (fourth-year) and Master's level. Also taught a Master's-level course at the University of Damascus (2008–2010).

ACADEMIC AND ADMINISTRATIVE RESPONSIBILITIES

Since Sep. 2022 **Fourth-Year Programme Coordinator**

Cybersecurity and Data Science *programme, ENSIBS*

Providing academic support to students; monitoring and recording grades and assessments related to their work placements.

Aug. 2017 – Aug. 2018 **Vice-Dean for Scientific Affairs**

Faculty of Computer Engineering, University of Homs

MASTER'S THESIS SUPERVISION

Supervised or co-supervised **8 Master's theses** at the Faculty of Computer Engineering, University of Homs (2014–2021), in cybersecurity, distributed systems, the Internet of Things (IoT), and machine learning.

RESEARCH INTERESTS AND ACTIVITIES (OVERVIEW)

My research focuses on network and distributed-systems security and, more recently, on artificial intelligence applied to cybersecurity. My work includes multicast routing security and error-correcting codes (PhD research at INRIA and France Télécom R&D); intrusion detection and web services security (University of Homs); and neuro-symbolic AI for fraud detection on the Ethereum blockchain, in collaboration with P.-F. Marteau (IRISA) (MLN 2025).

PUBLICATIONS

International Peer-Reviewed Conferences

IC1 Z. Khallouf, P.-F. Marteau. *Ensemble Neuro-Symbolic AI and Logic Tensor Networks for Detecting Fraud on the Ethereum Blockchain.* 8th International Conference on Machine Learning for Networking (MLN 2025), Paris, France, 2–4 December 2025. (In English)

IC2 Z. Khallouf. *Trust Management in Peer-to-Peer SIP Using the Security Assertion Markup Language.* International Conference on Information & Communication Technologies: From Theory to Applications (ICTTA'08), Damascus, Syria, 7–11 April 2008. (In English)

National Peer-Reviewed Conference

NC1 Z. Khallouf, V. Roca, R. Moignard, S. Loye. *A Filtering Approach for an IGMP Flooding Resilient Infrastructure.* 4th Conference on Network Security and Architecture (SAR'05), Batz-sur-Mer, France, 6–10 June 2005. (In English)

International Peer-Reviewed Workshop

W1 V. Roca, Z. Khallouf, J. Labouré. *Design and Evaluation of a Low-Density Generator Matrix (LDGM) Large Block FEC Codec.* 5th International Workshop on Networked Group Communication (NGC'03), Munich, Germany, September 2003. (In English)

National Journals (Syria)

Z. Khallouf, sole author. *Al-Baath University Journal.* Articles published in Arabic.

NJ1 *Classification of Intrusions in Local Area Networks (LANs) using Random Forests and Apache Spark*, 2021.

NJ2 *Securing Representational State Transfer (RESTful) Web Services Using OpenID Connect*, 2018.

NJ3 *Comparison between AngularJS 1 and Angular 2 in the Implementation of the Presentation Layer of Multitiered E-commerce Applications*, 2017.

NJ4 *Implementing Multi-Tiered E-commerce Applications and Assessing the Impact of the Different Techniques on Performance*, 2016.

NJ5 *Simulation of Routing Protocols in Mobile Ad hoc Networks (MANETs)*, 2011.

NJ6 *Survey and Taxonomy of Approaches to Improve the Efficiency of Intrusion Detection Systems*, 2010.

NJ7 *Studying the Use of Group Communication Algorithms to Build Fault-Tolerant Services via Replication*, 2010.

SELECTED PRESENTATIONS AND CONFERENCES

- MLN 2025, Paris, 2–4 Dec. 2025: presentation of IC1 (in English); Community Cyber UBS-Lorient, 4 Dec. 2025 (in English); Instant Cyber & AI 25, Université Bretagne Sud, May 2025 (in French).
- *Comparing Approaches to Improve the Efficiency of Intrusion Detection Systems*, International Conference on Cybercrime & Information Security (ICCIS 2010), Egypt (in English).

- *Data Protection Using Replication Based on Group Communication*, 6th ICT Security Forum, Damascus, Oct. 2010 (in English).
- *About Integrating Ethics in the Software Engineering Curriculum*, SCS Workshop, Damascus, Dec. 2007 (in Arabic).
- *Secure Multicast Routing Infrastructure: The Network Operator's Point of View*, SAR'04, La Londe, French Riviera, France, 21–25 June 2004 (in French).

LANGUAGES AND INTERESTS

Languages	Arabic, French, English (languages of instruction)
Art	Painting and sculpture: first solo exhibition in France (UBS, 2025); four group exhibitions in Vannes-France (2026)
Sports	UBS Relay Marathon (2023, 2024); Vannes Half Marathon (2026)

Last updated: November 2026